

COS-AT3 ASSESSMENT

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CSA0313

DATA STRUCTURE

1) Traverse an Array and Display elements

```
#include <stdio.h>
int main {
    int a[100], n, i;
    printf ("Enter number of elements :");
    scanf ("%d", &n);
    printf ("Enter elements : \n");
    for (i=0; i<n; i++)
        scanf ("%d", &a[i]);
    printf ("Array elements are : \n");
    for (i=0; i<n; i++)
        printf ("%d", a[i]);
    return 0;
}
```

2) Find the largest element in an array

```
#include <stdio.h>
int main () {
    int a[100], n, i, largest;
    printf ("Enter number of elements : ");
    scanf ("%d", &n);
    printf ("Enter element : \n");
    for (i=0; i<n; i++)
        scanf ("%d", &a[i]);
    largest = a[0];
    for (i=1; i<n; i++) {
        if (a[i] > largest)
            largest = a[i];
    }
}
```

```
printf ("largest element = %d", largest);  
return 0;
```

```
}
```

3) Find the Smallest Element in an Array.

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[100], n, i, smallest;
```

```
    printf ("Enter number of elements : ");
```

```
    scanf ("%d", &n);
```

```
    printf ("Enter elements : \n");
```

```
    for (i=0; i<n; i++)
```

```
        scanf ("%d", &a[i]);
```

```
    smallest = a[0];
```

```
    for (i=1; i<n; i++) {
```

```
        if (a[i] < smallest)
```

```
            smallest = a[i];
```

```
    }
```

```
    printf ("smallest element = %d", smallest);
```

```
    return 0;
```

```
}
```

4) Linear Search

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[10], n, i, key, found = 0;
```

```
    scanf ("%d", &n);
```

```
    for (i=0; i<n; i++) scanf ("%d", &a[i]);
```

```
    scanf ("%d", &key);
```

```
    for (i=0; i<n; i++)
```

```
        if (a[i] == key) { found = 1; break; }
```

```

if (found) printf ("found");
else printf ("Not found");
}

```

5) Binary Search

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[10], n, key, l=0, h, m, i;
```

```
    scanf ("%d", &n);
```

```
    for (i=0; i<n; i++) scanf ("%d", &a[i]);
```

```
    scanf ("%d", &key);
```

```
    h = n-1;
```

```
    while (l <= h) {
```

```
        m = (l+h)/2;
```

```
        if (a[m] == key) { printf ("found"); return 0; }
```

```
        if (a[m] < key) l = m+1;
```

```
        else h = m-1;
```

```
    }
```

```
    printf ("Not found");
```

```
}
```

6) Array Insertion

```
#include <stdio.h>
```

```
int main() {
```

```
    int a[10], n, p, x, i;
```

```
    scanf ("%d", &n);
```

```
    for (i=0; i<n; i++) scanf ("%d", &a[i]);
```

```
    scanf ("%d %d", &p, &x);
```

```
    for (i=n; i>=p; i--) a[i] = a[i-1];
```

```
a [p-1] = x;
```

```
n++;
```

```
for (i=0; i<n; i++) printf ("%d", a[i]);
```

```
}
```

7) Bubble sort

```
#include <stdio.h>
```

```
int main () {
```

```
int a[10], n, i, j, t;
```

```
scanf ("%d", &n);
```

```
for (i=0; i<n; i++) scanf ("%d", &a[i]);
```

```
for (i=0; i<n-1; i++)
```

```
for (j=0; j<n-1-i; j++)
```

```
if (a[j] > a[j+1]) {
```

```
t = a[j]; a[j] = a[j+1]; a[j+1] = t;
```

```
}
```

```
for (i=0; i<n; i++) printf ("%d", a[i]);
```

```
}
```

8) Selection sort

```
#include <stdio.h>
```

```
int main () {
```

```
int a[10], n, i, j, min, t;
```

```
scanf ("%d", &n);
```

```
for (i=0; i<n; i++) scanf ("%d", &a[i]);
```

```
for (i=0; i<n-1; i++) {
```

```
min = i;
```

```
for (j = i+1; j<n; j++)
```

```
if (a[j] < a[min]) min = j;
```

```
t = a[i]; a[i] = a[min]; a[min] = t;
```

```

}
for (i=0; i<n; i++) printf ("%d", a[i]);
}

```

9) Linked List Traversal

```

#include <stdio.h>
#include <stdlib.h>

```

```

struct node {int data; struct node * next;};

```

```

int main() {

```

```

    struct node * head = NULL, * p, * t;

```

```

    int n, i;

```

```

    scanf ("%d", &n);

```

```

    for (i=0; i<n; i++) {

```

```

        p = malloc (size of struct node);

```

```

        scanf ("%d", &p->data);

```

```

        p->next = NULL;

```

```

        if (head == NULL) head = p;

```

```

        else { t = head; while (t->next) t = t->next; t->next = p; }

```

```

    }

```

```

    for (t = head; t; t = t->next) printf ("%d", t->data);

```

```

}

```

10) BFS

```

#include <stdio.h>

```

```

int main() {

```

```

    int a[10][10], q[10], v[10] = {0};

```

```

    int n, s, f = 0, r = 0, i, j;

```

```

    scanf ("%d", &n);

```

```

    for (i=0; i<n; i++)

```

```

        for (j=0; j<n; j++) scanf ("%d", &a[i][j]);

```

```

    scanf ("%d", &s); s--;

```

$q[r++] = s; v[s] = 1;$

where $(f < n) \{$

$s = q[f++];$

$\text{printf} (" \%d", s, s++);$

$\text{for } (i = 0; i < n; i++)$

$\text{if } (a[s][i] \neq 0 \wedge !v[i]) \{$

$q[r++] = i; v[i] = 1;$

$\}$

$\}$

$\}$